

Warehouse management system for Southwestern Supply

Our team selected a supply chain and warehouse management system. The purpose of this system is to help the staff of Southwestern Supply to manage and lookup products, product locations, stock movement, staff accounts, customer information, supplier information, create and look up purchase orders, inbound deliveries, and outbound shipments. The users of this system will be the employees of Southwestern Supply. Each user will have certain roles attached which will give them different permissions.

Our entities will include the **product entity** which is responsible for storing information that each product will have. This includes the product number, name, category (such as apparel or electronics), unit cost, available inventory quantity, and assigned location number. The **product location** entity responsible for storing information related to the different locations within the warehouse for where product entities are located. Each product location stores a specific category of products that represents aisles and shelves with specific capacities (therefore the system will remember that they cannot retain more than what their capacity allows). The **stock movement receipt** entity is responsible for storing information related to when stock is moved from shelf to shelf or product location to product location within the warehouse. It will contain a unique receipt number, the list of products that were moved, their original location, their destination, the day they were moved, and the staff account number of who moved them. The **staff account** entity is responsible for storing information related to each employee in Southwestern Supply including account number, username, role (inbound management, outbound management, and stock management), and password. This will allow keeping track of which employees make which changes, so we can maintain staff accountability. The **customer entity** is responsible for storing information related to each customer. It will store the customer number, name, phone number, email, and shipping address of each customer. The **supplier** entity is responsible for storing information related to each supplier. It will store the supplier number, supplier name, supplier email, product types offered, and supplier phone number about each supplier. The **purchase order** entity is responsible for storing information about each purchase made to a supplier. It will store the supplier number, order number, date, total price, order status, staff account number, and product(s). The **inbound delivery** entity is responsible for storing the order number, delivery number, tracking number, carrier, arrival date, delivery status, and product(s) from an inbound delivery. The **purchase order** receipt is responsible for storing the order number, receipt number, payment type, date, total amount, and product(s). The **customer order** entity is responsible for storing the customer number, order number, date, total price, order status, staff account number, product(s) of a customer order. The **outbound shipment** is responsible for storing the customer number, receipt number, shipment number, tracking number, carrier, arrival date, shipment status, and product(s). Finally, the **outbound shipment receipt** is responsible for storing the customer number, receipt number, payment type, date, total amount, and product(s). These entities are important because they allow keeping track of all of the warehouse operations. They allow full management of internal stock, purchase orders coming

inbound to the warehouse, and shipments from customer orders out of the warehouse, along with keeping track of which employees do what. These entities are connected in many ways. For example, product is connected to product location because the items from the product entity must be stored inside a product location. Purchase order is connected to purchase order receipts, since after the order is added to the system the receipt must be created. The staff account entity is connected to the purchase order, customer order, and stock movement receipt. Firstly, depending on the role assigned to the employee, they will have access to one of those three functions (inbound management, outbound management, and stock management). When the information for these is filled out the staff number is part of the information required, allowing quick lookup of which employee made the order or moved the inventory.

An example of how this system can be used is an employee who is in charge of managing purchase orders using the system to check inventory and decide which items to order. Then, they can look at the list of suppliers and pick suppliers which offer the products they need to order. After they decide the suppliers and products they need to order, they can fill out a purchase order and attach their employee number. Once the order is delivered, based on the delivery date imported into the program, the stock will be updated. Another employee who is responsible for moving stock from a delivery to sections of storage will fill out a stock movement receipt with their employee ID attached. This will then keep a record of the stock movement and where each product has been moved to inside storage.

We may add additional entities such as a carrier entity to store more information about carriers or a product category entity to store more information about the different product categories.